Rishi Shah

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Education

McMaster University **Apr. 2029**
B.Eng. Computer Engineering | Schulich Leader Scholar ($120K, 1 of 50 in Canada) *Hamilton, Ontario*

Experience

E3A Healthcare **May 2026 – Aug 2026**
Machine Learning Engineer *Shenzhen, China*

- Set architecture direction for the flagship labor-prediction model, turning at-home fetal monitoring into a posterior over days to delivery: beating clinical SOTA by 72% for term delivery
- Designed the architecture the research team now builds on: cohort mixture-of-experts with a learned gate, quantile heads for calibrated uncertainty, a reasoning layer, and the sub-agent decomposition now on the roadmap
- Built the supervised layer and coupled-likelihood objective that halved the preterm MAE
- Authored the evaluation methodology, and automated multi-region GPU cluster provisioning on GCP (B200/H100), putting on-demand multi-node training in front of every engineer on the team

The Centre for Mechatronics and Hybrid Technologies **Oct. 2025 – Apr. 2026**
Research Engineer *Hamilton, ON*

- Cut battery voltage-model test RMSE 84% (12.82 mV to 2.03 mV) with a physics-ML hybrid (PyBaMM SPMe + seq2seq LSTM residual), then shipped it through ONNX to a real-time Simulink BMS block at bit-exact parity
- Implemented Extended Kalman Filter and SVSF state-of-charge estimators from scratch in NumPy with a genetic-algorithm 2RC-ECM parameter ID (0.58 mV mean HPPC RMSE); co-authored paper under submission

Western University **Apr. 2024 – Aug. 2025**
Machine Learning Researcher *London, ON*

- Architected a recurrent, denoising CORnet-S variant that beats MIT's CORnet-S on robustness with no adversarial training: learnable prefiltering, gated recurrence, and denoise-scaling reach 97.82% PGD on MNIST, 89.3% on CIFAR-100, and 84.7% on ImageNet100 at ∼15% extra compute

Robarts Research Institute **Nov. 2022 – May 2024**
Computer Vision Researcher *London, ON*

- Trained a U-Net with Synaptive Medical for surgical-tool segmentation on live OR video: 95% DICE, <0.12 s/frame

Publications

Rollout-Decoded Reconstruction for Long-Horizon Prediction in Latent World Models | **2026**
arXiv:2608.25017, cs.LG | first author

- One loss term that free-runs the model during training, decodes every rollout latent, and penalizes drift against ground truth: nearly doubles the horizon a latent world model stays accurate over, at zero added parameters and identical inference cost
- Raised valid prediction time 1.80x on chaotic Kuramoto–Sivashinsky (3.87 to 6.97 time units) at an identical 193,568 parameters, on seeds never used in selection, winning 10 of 10 preregistered configurations at 1.71–2.50x

A Contract-Grade Verifier for LLM-Generated GPU Kernels, and a Native Blackwell Backward for the Gated-Linear-Recurrence Family | *arXiv:2608.12700, cs.LG | first author* **2026**

- Wrote the first native Blackwell tcgen05/TMEM training backward for the gated-linear-recurrence family: one backward trains five architectures (LA, GLA, Mamba-2/SSD, KDA, Gated DeltaNet) at 3.3×10^{-3} worst relative error against an fp64 oracle, bit-for-bit deterministic, with a 1.1B-parameter model trained end-to-end through it
- Built a 12-gate contract verifier (tolerance-free gates, derived bounds, a 19-kernel red team) and audited 2,638 LLM-generated kernels a public system's own harness had accepted: 62.1% carry a violation, and the standard allclose check certifies 1,487 kernels the verifier rejects against 14 in the reverse direction

Projects

CGM-on-Chip | *C++17* **May 2026**

- Implemented a deep-learning stack in ∼5,500 lines of pure C++17 (from-scratch autograd, hand-rolled Adam, fork-join thread pool), fixed a box-projection clamp missing from OSDN's algorithm, and delivered 60-minute-ahead hypoglycemia prediction (AUROC 0.9113) on a $4 ESP32-S3 in 20% of its RAM, matching the FP64 trainer to 8×10^{-7}

Technical Skills

GPU & ML: CUDA, Triton, CUTLASS, PTX (tcgen05/TMEM), PyTorch, JAX, NumPy, ONNX
Languages & Tools: C++17, Python, TypeScript, MATLAB, SQL, Bash, GCP, Docker, Linux, Git